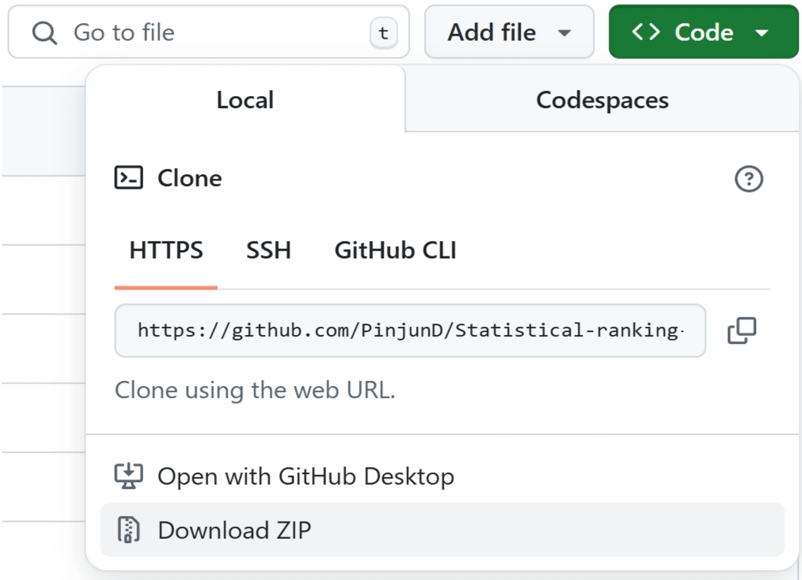
Guideline

1. Download the document

You can click the botton *Donwnload ZIP*:



If you have already installed *Git Bash*, you can use Git to clone the repository:

```
D:\>git clone https://github.com/PinjunD/Statistical-ranking-with-dynamic-covariate.git
Cloning into 'Statistical-ranking-with-dynamic-covariate'...
remote: Enumerating objects: 1124, done.
remote: Counting objects: 100% (382/382), done.
remote: Compressing objects: 100% (247/247), done.
remote: Total 1124 (delta 215), reused 266 (delta 130), pack-reused 742 (from 1)
Receiving objects: 100% (1124/1124), 129.41 MiB | 3.44 MiB/s, done.
Resolving deltas: 100% (660/660), done.

D:\>cd Statistical-ranking-with-dynamic-covariate

D:\Statistical-ranking-with-dynamic-covariate>
```

2.Confirm the Suitability of the Environment

Please access the Python environment and verify the package versions.

```
D:\Statistical-ranking-with-dynamic-covariate>python
Python 3.9.9 (tags/v3.9.9:ccb0e6a, Nov 15 2021, 18:08:50) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import numpy
>>> import pandas
>>> import matplotlib
>>> import joblib
>>>
>>> print(numpy.__version__)  # Required: at least 1.26.4
1.26.4
>>> print(pandas.__version__)  # Required: at least 2.2.1
2.2.1
>>> print(matplotlib.__version__)  # Required: at least 3.9.0
3.9.0
>>> print(joblib.__version__)  # Required: at least 1.4.2
1.4.2
>>> quit()
```

3. Simulation

- Please access the suitable path:

```
D:\Statistical-ranking-with-dynamic-covariate>cd Simulation

D:\Statistical-ranking-with-dynamic-covariate\Simulation>
```

- Please execute the following code in this path:

```
python "FigureS1.py" # In Simulation
```

- You will observe:

```
D:\Statistical-ranking-with-dynamic-covariate\Simulation>python FigureS1.py
In this program, we verify the uniform consistency of the MLE in the PlusDC model using simulated data.
We set n={200,400,600,800,1000} and conductthe simulations for NURHM and HSBM with 300 repeations.
Please note that the program may take a considerable amount of time to complete.
Start!
==========NURHM==========
----------
The 1st repetition of NURHM: number of nodes:200.
u_infty:0.4866537926674O716, v_infty:0.0066815124805834885.

----------
The 2nd repetition of NURHM: number of nodes:200.
u_infty:0.5115432850077104, v_infty:0.01771210625340358.

----------
The 3rd repetition of NURHM: number of nodes:200.
u_infty:0.6045637595370839, v_infty:0.011324221751249221.

----------
The 4th repetition of NURHM: number of nodes:200.
u_infty:0.6024801875842916, v_infty:0.03763539573611174.
```

4. Tennis Data

- Please access the suitable path:

```
D:\Statistical-ranking-with-dynamic-covariate\Simulation>cd..

D:\Statistical-ranking-with-dynamic-covariate>cd Tennis
```

- Please execute the following code in this path:

```
python "Figure5.py" # In Tennis
python "Figure1.py" # In Tennis
```

- You will observe:

```
D:\Statistical-ranking-with-dynamic-covariate\Tennis>python Figure5.py
In this program, we investigate the age effect among tennis players using PlusDC model.

==========Data Loading==========
Loading data from 'data\tennis(preprocessed).csv'...
Complete!

==========Data analysis==========
We construct covariates using Gaussian radial basis functions defined as:
f(t;a,ρ) = exp{-ρ(t - a)^2}, where parameters (a,ρ) ∈{25,30,35}×{0.01,0.03}
```

5. Horse-Racing Data

- Please access the suitable path:

```
D:\Statistical-ranking-with-dynamic-covariate\Tennis>cd..

D:\Statistical-ranking-with-dynamic-covariate>cd HorseRacing
```

- Please execute the following code in this path:

```
python "TableS2.py" # In HorseRacing
python "FigureS2.py" # In HorseRacing
python "FigureS3.py" # In HorseRacing
```

- You will observe:

```
D:\Statistical-ranking-with-dynamic-covariate\HorseRacing>python TableS2.py
In this program, we investigate the impact of covariates on horse racing results.
Specifically, we consider three factors to explain the internal scores of horses:
(1)Actual weight: The weight that a horse carries during a competition.
(2)Draw: The post position number assigned to a horse in a race.
(3)Public belief: The winning probability derived from the final win odds shortly before the competition begins.

=========Data Loading=========
To begin with, we load the data from 'data\runs(preprocessed).csv'...
|████████████████████████████████████████-| 100.0% Complete
```